$$
\begin{aligned}
V(x) &= E\big((X-x)^2\big) \\
&= E\big(((X-\mu)+(\mu-x))^2\big) \\
&= E\big((X-\mu)^2\big) + 2(\mu-x)E(X-\mu) + (\mu-x)^2 \\
&= \sigma^2 + 0 + (\mu-x)^2 \\
&= \sigma^2 + (x-\mu)^2
\end{aligned}
$$

$$
E(Y) = E\big(\sigma^2 + (X-\mu)^2\big) = \sigma^2 + E\big((X-\mu)^2\big) = \sigma^2 + \sigma^2
$$
$$
\implies \boxed{E(Y) = 2\sigma^2}
$$

$\because$ X is uniformly distributed on the interval $0 \le X \le 1$

$$
\therefore \mu_X = \frac{1}{2}, \qquad \sigma_X^2 = \frac{1}{12}
$$

$$
V(x) = \sigma^2 + (x-\mu)^2 = \frac{1}{12} + \left(x - \frac{1}{2}\right)^2 = \boxed{x^2 - x + \frac{1}{3}}
$$

$$
\begin{aligned}
F_Y(y) &= P(Y \le y) \\
&= P\left(\left(X - \frac{1}{2}\right)^2 + \frac{1}{12} \le y\right) \\
&= P\left(\frac{1}{2} - \sqrt{y - \frac{1}{12}} \le X \le \frac{1}{2} + \sqrt{y - \frac{1}{12}}\right)
\end{aligned}
$$

Since $0 \le X \le 1$,

$$
0 \le \left(X - \frac{1}{2}\right)^2 \le \frac{1}{4} \qquad \implies \qquad \frac{1}{12} \le Y \le \frac{1}{3}
$$

$$
F_X(x) = \begin{cases} 0 & x \le 0 \\ x & 0 < x < 1 \\ 1 & x \ge 1 \end{cases} \implies F_Y(y) = \begin{cases} 0 & y \le \frac{1}{12} \\ \sqrt{4y - \frac{1}{3}} & \frac{1}{12} < y < \frac{1}{3} \\ 1 & y \ge \frac{1}{3} \end{cases}
$$

$$
\begin{aligned}
F_Y(y) &= F_X\left(\frac{1}{2} + \sqrt{y - \frac{1}{12}}\right) - F_X\left(\frac{1}{2} - \sqrt{y - \frac{1}{12}}\right) \\
&= \left(\frac{1}{2} + \sqrt{y - \frac{1}{12}}\right) - \left(\frac{1}{2} - \sqrt{y - \frac{1}{12}}\right) \qquad \left(\frac{1}{12} \le y \le \frac{1}{3}\right) \\
&= 2\sqrt{y - \frac{1}{12}} \\
&= \sqrt{4y - \frac{1}{3}}
\end{aligned}
$$

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{2}{\sqrt{4y - \frac{1}{3}}} \qquad \left(\frac{1}{12} < y < \frac{1}{3}\right)$$

$$\implies \boxed{f_Y(y) = \begin{cases} 0 & otherwise \\ \frac{2}{\sqrt{4y-\frac{1}{3}}} & \frac{1}{12} < y < \frac{1}{3} \end{cases}}$$

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} y\, f_Y(y)\, \mathrm{d}y \\ &= \int_{1/12}^{1/3} 2y(4y - \frac{1}{3})^{-\frac{1}{2}}\, \mathrm{d}y \\ &= \left[y \left(4y - \frac{1}{3}\right)^{\frac{1}{2}} - \frac{1}{6}\left(4y - \frac{1}{3}\right)^{\frac{3}{2}} \right]_{1/12}^{1/3} \\ &= \frac{1}{6} \end{aligned}$$

From $(*)$, $E(Y) = 2\sigma^2 = 2 \cdot \frac{1}{12} = \frac{1}{6}$

This solution was submitted by **Jon**.

Published and refined by **Jon**.

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